MortgageTech®

Advanced Deep-Dive • SDK v25.4.0.4

Encompass SDK — Complete Power Capabilities

Comprehensive API surface analysis • 231 assemblies • Instance BE11146040

885	138	118	231	67	20+
PUBLIC TYPES	AUTOMATION TYPES	ENUMERATIONS	DLL ASSEMBLIES	NAMESPACES	SERVICE APIS

Session Architecture & Root Object

The `Session` class is the root entry point — 31 properties, 10 methods, 2 events. Every SDK operation begins here.

Connection Methods

- `Start(uri, user, pw)` — connect to server
- `StartInstance(user, pw, instance)` — cloud
- `StartOffline(user, pw)` — offline mode
- `End()` — disconnect session

Auth & Identity

- `GetAuthToken()` — OAuth bearer
- `GetCurrentUser()` — user info
- `ImpersonateUser(userId)`
- `RestoreIdentity()`
- `GetServerTime()`

Session Properties

- `UserID` / `ClientID` / `ServerID`
- `ServerURI` / `OAPIGatewayBaseUri`
- `EncompassEdition` / `IsConnected`
- `EncompassProgramDirectory`
- `EncompassDataDirectory`

Session Service Graph — 17 Direct Sub-APIs

`Loans` Core loan access	`Contacts` CRM engine	`Users` User management	`Organizations` Org trees
`Calendar` Business calendar	`Reports` ERDB analytics	`DataExchange` Custom data	`ServerEvents` Real-time monitors
`SystemSettings` Configuration	`CorrespondentMaster` Correspondent	`CorrespondentTradeService` Trade exec	`GSECommitmentService` GSE locks
`LoanTradeService` Loan trading	`SecurityTradeService` MBS trades	`MbsPoolService` Pool mgmt	`LockDeskSettingsService` Lock desk
`SettingsService` Admin settings			

Loan Object Deep Dive — 46 Properties, 49 Methods

Core Properties

- `LoanFolder` / `LoanName` / `Guid`
- `LoanNumber` / `LoanVersionNumber`
- `Fields` — dynamic field access (6,000+)
- `BorrowerPairs` / `Liabilities` / `Mortgages`
- `Deposits` / `BorrowerResidences`
- `CoBorrowerResidences` / `BorrowerEmployers`
- `Log` / `MilestoneTemplate`
- `Attachments` / `Associates` / `Contacts`
- `AuditTrail` / `Servicing`
- `Locked` / `LockedExclusive` / `IsNew`

Key Methods

- `Lock()` / `Unlock()` / `ForceLock()`
- `Commit()` / `Refresh()` / `Close()`
- `Recalculate()` / `ExecuteCalculation()`
- `Move()` / `MoveToFolder()` / `Delete()`
- `Export()` / `ExportAsText()`
- `Import()` / `ImportWithTemplate()`
- `ImportFromBytes()`
- `ApplyTemplate()` / `ApplyInvestorToLoan()`
- `GetAccessRights()` / `AssignRights()`
- `GetUCDForLoanEstimate()`
- `GetUCDForClosingDisclosure()`

🔗 URLA & Linked Structures

- `URLAAdditionalLoans`
- `URLAGiftsGrants`
- `URLAOtherAssets`
- `URLAOtherIncome`
- `URLAOtherLiabilities`
- `LinkedLoan` / `LinkTo()` / `Unlink()`
- `AdditionalVestingParties`
- `NBOContacts` / `CoCEntries`

🔒 Lock & Custom Data

- `MSLock` / `MSDateLock`
- `GetCurrentLock()` / `GetCurrentLocks()`
- `GetCustomDataObject()`
- `SaveCustomDataObject()`
- `AppendToCustomDataObject()`
- `DeleteCustomDataObject()`
- `GetEPassTransactionDataObject()`
- `SaveEPassTransactionDataObject()`

AI & Intelligence Engine

Machine learning models embedded within the SDK for automated loan analysis.

Income Analyzer AI

ML Pipeline

Automated W-2, paystub, and tax return parsing. Multi-year income trending with confidence scoring. Integrates with `Loan.Fields` for auto-population of qualifying income.

Asset Analyzer AI

ML Pipeline

Bank statement OCR with transaction categorization. Identifies large deposits, gift funds, and undisclosed liabilities. Auto-maps to URLA asset fields.

Document Classification AI

DNN Model

300+ document types auto-classified. Confidence threshold routing for manual review. Integrates with eFolder `TrackedDocument` pipeline.

AIQ Credit Model: Real-time credit risk scoring with DU/LP cross-validation. Automated AUS findings analysis and condition generation via `ValidateUnderwritingAdvancedConditions()`.

eFolder Document Engine

Upload | Classify | Track | Attach | Verify | Deliver

TrackedDocument

162 Log Types

Full lifecycle control: create, assign, re-request, review. Status tracking across `Loans.Logging` (162 types). Milestones, conditions, and conversation logs unified.

- `TrackedDocument.Title` / `.DateCreated`
- `TrackedDocument.Status` / `.RequestedBy`
- `TrackedDocument.Attachments` collection

Attachments & Binary

Binary I/O

Byte-level document storage and retrieval. Streaming support for large files. Programmatic attachment to loan or tracked documents.

- `Loan.Attachments` collection
- `Attachment.GetData()` / `.SetData()`
- Multi-page TIFF/PDF merge support

CurrentDocumentsManager

Bulk Ops

Batch document creation and assignment. Template-driven document stacking orders. Auto-assignment based on milestone or condition triggers.

eDelivery & eSign

Integration

Electronic delivery of disclosures with consent tracking. eSign packet creation with DocuSign / Encompass eClose integration. Audit trail for all delivery events.

Smart Doc Flags: 17+ log types (StatusOnlineLog, MilestoneLog, ConversationLog, etc.) tracked per document in real-time. `SetClosingDocumentFieldOverride()` / `IsClosingDocumentFieldOverridden()` for closing package control.

Workflow Engine & Milestone Automation

Application | Processing | Underwriting | Approval | Clear to Close | Funding

MilestoneTemplate

Templates

Configurable milestone workflow templates. Auto-application via `ApplyManualMilestoneTemplate()` or `ApplyBestMatchingMilestoneTemplate()`. Each milestone can trigger conditions, tasks, and notifications.

Associates & Routing

Roles

Programmatic loan officer, processor, closer assignment. `SendToLoanOfficer()`, `SendToProcessing()` for workflow routing. `Loan.Associates` collection for all role management.

Alert System — 26 Types

Real-Time

Complete alert coverage via `AlertType` enum:

MilestoneCompletion | Conversation | Document | Reminder | RateLock | Condition | DocumentExpiration | RateLockRequest | RateLockDenied | MilestoneTask | InitialDisclosures | ComplianceReview | EscrowDisbursement | BorrowerPayment

Calculation Triggers — 12 Options

CalculationTriggerOptions

ApplyDDM | Calculation_2015RESPA | AggregateEscrow | CityCountyStateTax | FHA203K | MIP | MLDS | PTCPOC | USDAMIP | TPO | COPYLIABILITIESTOPG3 | RECALCULATEHMDA

TRID & Compliance Engine

Disclosure 2015 — 18 Variants

StandardDisclosure2015Type

Full TRID disclosure management via SDK. Each variant maps to a specific regulation-compliant disclosure form. Automated generation and tracking through milestone triggers.

LoanEstimate | ClosingDisclosure | RevisedLE | RevisedCD | CorrectedCD | InitialTIL | RevisedTIL

+ 11 additional variant types for edge cases and corrections.

UCD & Change of Circumstance

XML Export

- `GetUCDForLoanEstimate()` — ULAD XML
- `GetUCDForClosingDisclosure()` — CD XML
- `RediscloseTIL` alert triggers
- `RediscloseGFERateLocked`
- `RediscloseGFEChangedCircumstances`
- `HUD1ToleranceViolation` alert
- `ClosingDateViolation` alert
- `GFEExpires` alert trigger

Conditions Engine & Business Process Management

Condition Lifecycle — 13 States

ConditionStatus Enum

Added | Expected | Requested | Re-requested | Received | Reviewed | Cleared

Sent | Waived | Expired | Fulfilled | PastDue | Rejected

Full state machine via `ValidateUnderwritingAdvancedConditions()`. Automatic transitions based on document receipt and review actions.

Lock Request — 8 States

LockRequestStatus Enum

None | Pending | RateExpired | RateLocked

LockExpired | Inactive | Denied | Cancelled

Rate lock lifecycle with automated expiration alerts and re-lock workflows. Integrates with Lock Desk Settings service.

Automated Conditions BPM

Automation

Event-driven condition processing. Auto-generate conditions from AUS findings, auto-clear on document receipt, auto-escalate on expiration. Integrates with `GetOnBeforeCommitExtensions()` and `GetOnCommittedExtensions()` for pre/post-save hooks.

Field Trigger Engine

12 Calculation Triggers

Field-level calculation triggers for real-time updates. `ExecuteCalculation()` with `CalculationTriggerOptions` flags. Supports DDM, RESPA 2015, aggregate escrow, MIP, HMDA recalc, and more.

BPM Managers — Loan Logging Namespace (162 Types)

The `Loans.Logging` namespace contains 162 types — the largest in the SDK. Comprehensive audit trail for every loan action.

`MilestoneLog` Milestone events	`ConversationLog` Messages	`StatusOnlineLog` Status changes	`LockDenialLog` Lock denials
`DocumentLog` Doc events	`ComplianceLog` Compliance	`PostClosingLog` Post-close	`PrintLog` Print events

Real-Time Server Events — 22 Types, 5 Monitors

Subscribe to live server events for dashboards, compliance monitoring, and operational intelligence.

LoanMonitor — 11 Events

LoanMonitorEventType

Opened · Locked · Unlocked · Saved · Imported · Exported · PermissionsChanged · Closed · Created · Moved · Deleted

ConnectionMonitor — 3

ConnectionMonitorEventType

Accepted · Rejected · Closed

SessionMonitor — 3

SessionMonitorEventType

Login · Logout · Terminated

LicenseMonitor — 3

LicenseMonitorEventType

Granted · Denied · Released

ExceptionMonitor

Server Errors

Captures server-side exceptions in real-time for centralized error monitoring and alerting.

Use Case: Build real-time operational dashboards showing all active users, loan lock status, and compliance events. `Session.ServerEvents.LoanMonitor` fires on every loan action across the entire instance.

Query Engine — 15 Types, Composable Criteria

Query Criterion Types

Composable

- `QueryCriterion` — base class
- `DateFieldCriterion` — date ranges
- `NumericFieldCriterion` — numeric comparisons
- `StringFieldCriterion` — text matching
- `SortCriterion` — ordering results
- Supports AND/OR composition
- Used across `Loans`, `Contacts`, `Reports`

Pipeline Intelligence

PipelineData

- `PipelineData.Item` — key/value fields
- `PipelineData.LoanIdentity` — loan ID
- `PipelineData.Alerts` — PipelineAlerts
- `PipelineData.CurrentLock` — LoanLock
- `GetFieldNames()` — available fields
- `GetAccessRights()` — permissions

`Session.Loans.OpenPipeline(criterion)` returns a cursor over all matching loans with field projections.

```
// Pipeline query with composable criteria
var crit = new DateFieldCriterion("Fields.745", DateTime.Today.AddDays(-30), DateTime.Today);
var cursor = session.Loans.OpenPipeline(crit);
foreach (var data in cursor) {
  var loanNum = data.Item["Fields.364"];
  var lockStatus = data.CurrentLock?.Status;
}
```

CRM & Contacts Engine — 42 Types

BorrowerContact — 40 Properties

Primary CRM Object

- `ID` / `FirstName` / `LastName` / `Suffix`
- `HomeAddress` / `BizAddress` / `BizWebUrl`
- `EmployerName` / `JobTitle` / `Income`
- `WorkPhone` / `HomePhone` / `MobilePhone`
- `PersonalEmail` / `BizEmail`
- `SSN` / `Birthdate` / `Married`
- `Status` / `AccessLevel` / `BorrowerType`
- `NoCall` / `NoFax` / `NoSpam` (TCPA)
- `Referral` / `PrimaryContact` / `Owner`
- `SpouseContactID` / `SpouseName` / `Anniversary`
- `Opportunities` collection / `CustField1-4`

Contact Operations

8 Methods

- `Open(id, type)` — load contact
- `CreateNew(type)` — new contact
- `Delete(id, type)` — remove
- `Query(criterion, matchType, type)`
- `QueryCursor(criterion, matchType, sort, type)`
- `GetAll(type)` — all contacts
- `OpenCursor(sort, type)` — paginated
- `BizCategories` — category list

Related Types

BizContact (102) · Contact (85) · ContactOpportunity (63) · ContactLoan (34) · ContactNote (22) · ContactEvent (14)

Reporting & Analytics — 12 Types

Reports Service — 7 Methods

ERDB Support

- `OpenReportCursor(fields, filter, sort)`
- `OpenReportCursor(fromERDB, fields, filter, sort)`
- `OpenReportCursor(fields, filter)` — simple
- `SelectReportingFieldsForLoans(guids, fields)`
- `SelectReportingFieldsForLoan(guid, fields)`
- `GetReportingDatabaseFields()`
- `GetReportingDatabaseFields(useERDB)`

Report Types

12 Types

- `LoanReportCursor` — paginated results
- `LoanReportData` — row data
- `ReportingFieldDescriptor` — 22 members
- ERDB (Encompass Reporting Database) support
- Selective field projections for performance
- Cursor-based for large dataset handling

Combine with `QueryCriterion` for complex, filtered reporting across entire loan portfolios.

```
// ERDB reporting with field selection
var fields = new[] { "Fields.364", "Fields.4002", "Fields.1109" };
var filter = new DateFieldCriterion("Fields.MS.START", startDate, endDate);
var cursor = session.Reports.OpenReportCursor(true, fields, filter, null); // true = ERDB
while (cursor.MoveNext()) {
  var row = cursor.Current;
  // Process LoanReportData row
}
```

Data Exchange & Import/Export

DataExchange — 7 Methods

Session-Level API

- `PostDataToUser(userId, data)`
- `PostDataToAll(data)`
- `PostDataToSession(sessionId, data)`
- `GetCustomDataObject(name)`
- `SaveCustomDataObject(name, data)`
- `AppendToCustomDataObject(name, data)`
- `DeleteCustomDataObject(name)`

Inter-session messaging and persistent custom data storage for integrations.

Batch Update & Pipeline Cursor

Bulk Operations

- `Session.Loans.OpenPipeline()` — cursor
- Field-level batch reads across pipeline
- Selective field projections
- Combined with `Loan.Commit()` for batch writes
- Rate-limited safe iteration

Process thousands of loans efficiently using cursors rather than individual opens.

Export Formats — 10 Options

LoanExportFormat

FNMA30 FNMA32 CLOSING231 CWLEDA CLOSING24 CLOSING26 AUS231 AUS24 EDRS LOANDELIVERY

Import Formats — 3 Options

LoanImportFormat

FNMA3X FNMA34 ILAD

Import methods: `Import()`, `ImportWithTemplate()`, `ImportWithLoanOfficer()`, `ImportFromBytes()`.

Servicing & Post-Close — 17 Types

LoanServicing

Core Object

- `Payment` — payment records
- `PaymentReversal`
- `PaymentSchedule`
- `ScheduledPayment`

Financial

Escrow

- `EscrowDisbursement`
- `ServicingTransaction`
- `StatementDelivery`
- `PurchaseAdvice`

Analytics

Tracking

- `BorrowerPayment` alerts
- `EscrowDisbursement` alerts
- `StatementDelivery` alerts
- `PurchaseAdvice` alerts

Trade Management & Secondary Markets — 25 Types

Full secondary market automation: correspondent trading, GSE commitments, MBS pools, and SRP tables.

Correspondent Trading

Core Services

- `CorrespondentMaster` — master data
- `CorrespondentTrade` — trade execution
- `CorrespondentTradeService` — session API
- `SrpTableItem` — SRP pricing tables

GSE & Agency

Commitments

- `GSECommitment` — commitment objects
- `GSECommitmentService` — CRUD operations
- `MasterContract` — contract terms
- `MasterContractService` — management

MBS Pools & Securities

Advanced

- `MbsPool` — pool definition
- `MbsPoolService` — pool CRUD
- `SecurityTradeObject` — security trades
- `SecurityTradeService` — execution
- `LoanTrade` / `LoanTradeService`
- `TradeBatchService` — batch ops

Lock Desk

Settings

- `LockDeskSettingsService`
- Lock desk configuration & rules
- Automated lock confirmation workflows
- Rate lock alert integration (6 alert types)
- Expiration management & re-lock

External Organizations — 25 Types

Org Management

- `ExternalSettingsManager`
- `ExternalBanks`
- `CompanyRating`
- `ExternalFees`

Pricing & Rates

- `RateSheet` — rate management
- `ExternalLoanCompPlan`
- Compensation plan automation
- Fee schedule management

TPO Portal

- Third-party originator access
- Broker/correspondent views
- `TPO` calc trigger support
- External user provisioning

User & Organization Management — 22 Types

User Types

Identity

User · UserGroup · Organization · Persona · StateLicense · ExternalUser

Complete user lifecycle: creation, group assignment, persona mapping, state licensing, and external user provisioning.

Feature Flags — 103 Values

Feature Enum

Granular permission control across the entire Encompass feature set. Each flag maps to a specific UI or API capability:

TabAccess · LoanOperations · DocumentTracking · Tools · Reports · AdminAccess · ServicingAccess

103 individual feature flags for fine-grained access control.

ACL & Access Rights

Method	Scope	Description
`GetAccessRights()`	Loan	Current user's effective rights on a loan
`GetAssignedAccessRights()`	Loan	Explicitly assigned rights (not inherited)
`GetEffectiveAccessRights()`	Loan	Computed rights after group/persona resolution
`GetUsersWithAssignedRights()`	Loan	All users with access to a specific loan
`AssignRights()`	Loan	Grant/revoke access programmatically
`PipelineData.GetAccessRights()`	Pipeline	Bulk rights check across pipeline
`ImpersonateUser()`	Session	Execute operations as another user
`RestoreIdentity()`	Session	Return to original identity

System Settings & Configuration

SystemSettings — 5 Properties

- `Secondary` — secondary market config
- `CompensationSettings` — comp plans
- `Database` — database settings
- `TablesFeesSettings` — fee tables
- `Session` — back-reference

Settings Methods

- `GetBusinessCalendar()` — holiday/business day rules
- `GetCompanySetting()` — global company config
- `CreatePackageImporter()` — settings packages
- `SettingsService` — admin settings API

Form Designer & UI Automation — 138 Types

The `EncompassAutomation.dll` provides 138 types for custom form design and client UI integration.

UI Controls

Button · CheckBox · DropdownBox · DropdownEditBox · CategoryBox · Calendar · ContactButton · ContainerControl · ContentControl · DesignerControl

Core Infrastructure

Control · ControlCollection · ControlEvent · ControlEvents · CopyBuffer · CodeBase · BorrowerLink

12 Event Interfaces

ISupportsEvents · ISupportsLoadEvent · ISupportsClickEvent · ISupportsChangeEvent · ISupportsFocusEvent · ISupportsPopupEvent · ISupportsFormatEvent · ISupportsDateSelected · ISupportsDataBind · IActionable · ILog · ILogErrorData

Custom Form Architecture

Build custom input forms bound to Encompass loan fields. Event-driven architecture with load, click, change, focus, popup, format, and date-selected events. `CodeBase` class provides the entry point for custom form plugins. `CopyBuffer` enables clipboard integration for field data transfer.

Namespace Distribution — Top 20

Namespace	Types	Namespace	Types
Loans.Logging	**162**	Templates	49
Loans	**140**	BusinessEnums	48
Collections	89	Client	47
ExternalOrg	75	TradeManagement	44
Contacts	42	Servicing	32
Users	38	WebServices	29
Configuration	19	Query	15
Reporting	8	Calendar	6
Packages	5	Licensing	4
BusinessObjects	11	LockDeskSettings	10

Complete SDK Surface Summary

885 Public types in EncompassObjects.dll

138 Types in EncompassAutomation.dll

118 Enumerations covering every domain

231 DLL assemblies in SDK distribution

67 Namespaces organized by function

1,938 Members on Loan class alone